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Cybersecurity Performance of Shanghai Cooperation Organization Countries and Türkiye Using the Hybrid MEREC-MAUT Approach

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Abstract: *Besides the benefits, digitalization has brought certain responsibilities and risks into everyday life. Cyberattacks, which are hard to predict, can cause serious harm to individuals, institutions, and countries. As a result, raising awareness about cybersecurity and assessing cybersecurity performance have become crucial. These threats, which have grown from regional to global issues, demand an analysis of the cybersecurity policies of international organizations and the capabilities of member states. This study aims to address a gap in regional cybersecurity evaluations by examining Shanghai Cooperation Organization (SCO) member states alongside Türkiye, considering its strategic role as a dialogue partner. For this purpose, it introduces a novel hybrid MEREC–MAUT framework to offer a structured comparative assessment. The 2024 Global Cybersecurity Index (GCI) data were analyzed using a hybrid approach that combines MEREC (Method based on the Removal Effects of Criteria) and MAUT (Multi-Attribute Utility Theory). MEREC was selected for objective weighting based on criteria removal effects, while MAUT offers flexible utility-based ranking, providing a robust analytical approach over traditional multi-criteria decision making (MCDM) methods. The analysis indicated that 'technical measures' ranked as the top criterion, followed by capacity-development and organizational measures. Türkiye, India, and Pakistan emerged as the top-performing countries, whereas Tajikistan and Belarus exhibited the lowest performance by a significant margin. The empirical results highlight the critical importance of regional collaboration within the SCO and reveal that prioritizing technical measures is essential for boosting cybersecurity performance. Thus providing policy-makers with a strategic improvement roadmap for targeted infrastructure updates in underperforming nations.*

Keywords: Cybersecurity, SCO Countries, Global Cybersecurity Index, MEREC, MAUT

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1. Introduction

The historical roots of digitalization show that it gained momentum in the late 1970s and 1980s, driven by the widespread adoption of personal computers. In particular, the civilian adoption and global growth of the internet after the Cold War marked a major turning point. Subsequently, the rise of smartphones and recent advances in artificial intelligence have made digitalization an essential part of everyday life. During this process, governments began digitalizing the management and control of public services considered critical infrastructure by incorporating innovations in network technologies. As a natural

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extension of this change, securing the cybersecurity of this critical infrastructure has become a top national security and strategic priority for modern states (Darıcılı, 2019).

The International Organization for Standardization (ISO) defines cybersecurity as 'the preservation of confidentiality, integrity and availability of information in the Cyberspace (Hathaway & Klimburg, 2012). According to the United Nations International Telecommunication Union (ITU), a leading and long-standing agency in digital technologies (ICT), cybersecurity involves various tools, policies, security concepts, and measures. It also encompasses guidelines, risk management approaches, actions, training, and best practices. Furthermore, it includes assurance and technologies used to protect the cyber environment and the assets of organizations and users (ITU, 2009). Cyberattack is defined as planned and coordinated assaults on the information systems or communication infrastructures of targeted individuals, corporations, institutions, and organizations. These attacks can be executed for commercial, political, or military purposes. When these attacks target a country, they are referred to as cyber warfare. Cyber-attacks can be carried out using various methods such as data diddling, salami techniques, super zapping, Trojan horse, or malware like viruses, logic bombs, phishing, chameleon, spam, scavenging, masquerading, hacking, illegal content, website theft and redirection, and social engineering (Aslay, 2017). Various global measurement methods have been developed to assess the defense performance of states and institutions against these complex cyber threats. Therefore, Cisco Cybersecurity Readiness Index, National Cybersecurity Index (NCSI), European Union Agency for Cybersecurity, and Global Cybersecurity Index (GCI) are used to evaluate the cybersecurity levels of countries and organizations. Among these, the GCI is recognized as a prominent, comprehensive, and widely utilized metric.

Cybersecurity has evolved into a critical threat factor on a global scale. It spans multiple dimensions, including economic, humanitarian, and national security concerns. From an economic perspective, malicious cyber activities are observed to inflict substantial financial damages across various countries, most notably in the United States (US). On an individual level, the loss of access to and control over confidential data has transformed into a normalized phenomenon among users. Furthermore, high-profile cyber incidents clearly demonstrate the critical importance of cybersecurity for national sovereignty and state security. Prominent examples include the deployment of the Stuxnet worm, Russian interference in elections, and the targeting of a nuclear power plant in India (Ruhl et al., 2022).

It is anticipated that future cyberattacks will become significantly more dangerous than they are today. Critical infrastructures (CIs) form the bedrock of societal well-being, economic stability, and national security. They encompass vital sectors such as energy grids, financial systems, telecommunications, transportation, healthcare, and water management. Today, these systems are adapting at an unprecedented pace to next-generation technological paradigm shifts. These shifts include cloud computing, virtualization technologies, Internet of Things (IoT) integration, and 5G networks. However, this digital transformation process increasingly exposes critical infrastructures to a growing array of cybersecurity risks. These risks are highly sophisticated, dynamic, and extend far beyond traditional threats (Nemure et al., 2026). Attacks organized by hostile adversaries against Industrial Control Systems (ICS), which provide vital services to critical infrastructures such as communication, manufacturing, and energy, pose a serious threat to the daily functioning of nation-states. ICSs have unique performance and reliability requirements. These requirements typically prioritize availability and integrity, followed by confidentiality. If not managed correctly, they pose significant risks to human health and safety, environmental protection, and financial stability through production losses. The unavailability of critical infrastructure (e.g., electrical power, transportation) can lead to economic impacts far beyond the systems directly and physically damaged. These impacts can adversely affect the local, regional, national, or potentially global economy (Maglaras et al., 2018).

Consequently, it is imperative to protect all actors, from the individual to the state level, against risks such as online fraud, identity theft, and the malicious activities of various terrorist groups. Furthermore, ensuring the privacy of personal data and financial information, as well as safeguarding intellectual and industrial property rights, remains essential. In this context, proactive measures taken by national and local governments, along with the immediate neutralization of potential attacks, will also prevent severe reputational damage. The exponential growth of data volume in the digital ecosystem implies that the

amount of data requiring protection will increase analogously. This trend indicates that the budget share allocated by governments to cybersecurity expenditures will progressively rise. According to a study commissioned by NATO, global costs resulting from cyber insecurity are projected to reach approximately 90 trillion $ by 2030 (Aslay, 2017). Moreover, it is argued that the future financial burdens induced by cyber insecurity may outweigh the benefits brought by cyberspace itself. Therefore, the expenditure required for robust cybersecurity measures is predicted to remain remarkably low compared to the total potential costs that insecurity could incur (Aslay, 2017; Goutam, 2015). In this context, one of the main goals of the proposed modern security solutions is to develop new methods that can detect and disrupt the activities of attackers after they gain access to the system. Special importance should be given to the implementation of new strategies that can detect, prevent, and mitigate data exfiltration attacks. This is because attack detection/prevention strategies are now considered insufficient for data protection (Maglaras et al., 2018).

These escalating global cyber threats and massive financial risks have made it difficult for nation-states to combat them single-handedly. Consequently, regional and international collaborations are necessitated to ensure security. In this context, the role of regional organizations is steadily expanding in order to execute cybersecurity strategies jointly and to enhance the capacities of member states. Among these, the SCO has become a global power hub geopolitically and economically in the Eurasian region. Consequently, it stands out with respect to the cybersecurity infrastructures and digital resilience of its member states.

In response to this requirement, measuring the effectiveness of cybersecurity policies and evaluating national performances through objective criteria constitute the core motivation of this study. The present study aimed to introduce the GCI as an important cybersecurity metric and conduct an empirical analysis of the cybersecurity performance of SCO member countries and Türkiye. Including Türkiye in this evaluation appears suitable given its status as an official SCO Dialogue Partner and its relatively advanced digital infrastructure. Furthermore, its established cybersecurity capacity may provide informative benchmarking insights for member states. Through this analysis, it seeks to identify their respective strengths and weaknesses. Therefore, GCI pillar data were analyzed using an integration of the MEREC and MAUT methods, both current MCDM approaches. A review of cybersecurity literature revealed no studies applying the MEREC-MAUT hybrid approach to SCO countries. In this hybrid framework, MEREC is utilized for objective criterion weighting based on criteria removal effects. Meanwhile, MAUT is employed for performance ranking through utility functions. By integrating these two methods, the study combines data-based objectivity with benefit-oriented analytical flexibility. This transparent and mathematically verifiable model offers an effective framework for measuring cybersecurity performance, with its detailed methodological rationale elaborated in the Methodology section.

The present study consists of five sections. Following the introduction, the literature review covers studies on cybersecurity and MCDM methods. The third section describes the application of the MEREC and MAUT methods. The fourth section presents and interprets the analysis findings. In the conclusion, the key findings are summarized in relation to the literature, along with the study's limitations, recommendations for future research, and policy suggestions.

2. Literature Review

This section reviewed empirical and methodological studies that contributed to the cybersecurity literature. Initially, empirical studies on cybersecurity based on international indices such as the GCI and NCSI are discussed. Among various global cybersecurity frameworks, the literature review specifically focuses on the GCI and NCSI due to their comprehensive coverage and standardized metrics. Furthermore, these indices offer widespread academic acceptance and directly align with the empirical focus of this study. Then, the applications of the MEREC and MAUT methods in the literature, which serve as the methodological framework of the current study, are reviewed. The literature review revealed that cybersecurity has been studied extensively due to its broad, multidimensional nature. However, most studies focus on technical, legal, and theoretical frameworks; studies on GCI performance analysis are relatively limited.

First, regarding index-based performance evaluations, empirical studies have employed MCDM models and statistical techniques to analyze GCI and NCSI data. Exploring the influence of national culture on cybersecurity, Onumo et al. (2017) utilized hierarchical regression to examine the relationship between GCI and Hofstede's cultural dimensions across 89 countries. The results indicated that individualism emerged as a major cultural factor influencing GCI. In another study by Altıntaş (2022), using the Entropy-based MABAC hybrid model, the cybersecurity performance of the G7 countries was ranked as the USA, England, Germany, France, Canada, Japan, and Italy. Focusing on the structural dynamics of the GCI, Bruggemann et al. (2022) analyzed the ranking effects of the five basic components of the GCI with Hasse diagrams, based on data from the top 11 countries in the GCI 2020 ranking. Their findings indicated that the technical component played a primary role in determining the rankings was technical, followed by capacity building and cooperation. Additionally, Singapore and Malaysia were found to be completely isolated from other countries. Overall, these empirical findings tend to highlight the versatility of index data across different analytical frameworks. However, a methodological distinction appears to exist among them: while statistical models primarily focus on external factors like culture, MCDM frameworks lean toward examining the internal components and ranking structures of the indices.

Building upon index rankings, recent literature increasingly expands the analytical scope toward efficiency and development indicators. Küpeli and Söğüt (2024) applied data envelopment analysis (DEA) using the global uncertainty index (GUI) and internet user (IU) data as input variables, with GCI, NCSI, and digital development level (DDL) as output variables. In a study on the cybersecurity efficiency of G20 countries, Indonesia, France, India, and Japan were identified as efficient, while Mexico underperformed. Similarly addressing indicator effectiveness, Prasetya et al. (2024) assessed NCSI data from 10 Southeast Asian countries using regression analysis. Using quadratic, cubic, logarithmic, S, logistic, and exponential equations, they found that policy development, cyber threat analysis and information, protection of essential services, and anti-cybercrime measures were effective indicators. In parallel, Fagarazzi (2024) studied the correlation between DDL and NCSI across 155 countries and found a strong positive relationship between the two. Research indicates that countries with higher levels of digital development are better at securing their digital infrastructure. Methodologically, these findings imply that higher digital infrastructure alone may not necessarily guarantee superior performance without adequate operational efficiency and targeted policy measures.

Recent studies appear to shift the focus from global frameworks to targeted regional dynamics. They highlight the potential impact of geopolitical alliances and crises on index performance. Öztaş (2025) evaluated BRICS nations with the integrated LOPCOW–MARCOS approach, concluding that cooperation measures carry the highest weight. In the performance rankings, the United Arab Emirates, Indonesia, and Egypt were reported to achieve the highest performance scores. Focusing on geopolitical transitions, Öztaş (2026) analyzed NCSI data for EU candidate countries using the Centroidous-RAPS method during pre- and post-pandemic periods. The analysis showed that policy development, incident response, and cybersecurity training became more important after the pandemic. It also revealed how countries prepared their cybersecurity defenses amid the accelerated digitalization caused by the pandemic. Findings showed that Georgia and Ukraine turned the crisis into an opportunity by strengthening their cybersecurity capacities strategically, while Türkiye showed moderate performance. In general, these evaluations suggest that MCDM methodologies offer noticeable adaptability. They tend to capture how regional integration and external shocks reshape cybersecurity priorities over time.

Second, regarding objective weighting methodologies, recent literature points toward the theoretical evolution and cross-domain applicability of the MEREC method. Keshavarz-Ghorabaee (2021) introduced the MEREC method, which is based on the removal of criteria from the system, and conducted a comprehensive simulation-based analysis to verify its effectiveness and stability. The analysis showed that the MEREC method produces highly stable, reliable, and consistent results compared to other objective weighting methods. Owing to these advantages, Simic et al. (2022) conducted a study using a hybrid model that integrates the MEREC and MARCOS methods in a Type-2 neutrosophic (T2NN) environment to reduce the climate effects of urban transportation. The research findings indicated that land-use planning emerged as a

highly effective sustainable policy. The study also emphasized that municipalities could generate additional short-term income by implementing low-emission zones and limited traffic zones. Similarly, Ecer and Ayçin (2023) analyzed the innovation performance of G7 countries using seven MCDM methods and employed the MEREC method to determine the criterion weights. The results revealed that business sophistication emerged as the primary indicator in innovation performance, with the USA ranking first. Focusing on sensor selection in underground mining operations, Wang et al. (2024) examined 15 criteria and 10 sensor technologies using the MEREC-CoCoSo integrated method. The findings showed that ease of installation ranked as the top criterion for sensor selection. When prioritizing alternative sensor technologies, the results indicated that the top-ranked option for underground mine safety was proximity sensors, followed by biometric, gas, and temperature and humidity sensors. Furthermore, Ali (2025) proposed a new integrated model combining LqROF values based on Archimedean copula and co-copula (ACC) theory with the MEREC weighting technique to address plastic waste management issues. The study, supported by a realistic case analysis and comparative evaluations, demonstrated that the proposed method delivers reliable, stable results in ecological decision-making under conditions of imprecision and uncertainty. Taken together, these studies tend to support the reliability of MEREC across diverse fields. They suggest its potential adaptability when integrated with various hybrid models.

Third, regarding utility-based ranking frameworks, recent literature draws attention to the operational flexibility and precision of the MAUT approach. Kim and Song (2009) used the MAUT approach to choose the best dismantling scenario in the thermal column dismantling plan for Korea Research Reactor-1 (KRR-1). The study combined key criteria, including dismantling cost, schedule, and radiation safety, showing the usefulness of the MAUT method in nuclear engineering projects. Demonstrating its applicability in organizational decision-making, Taufik *et al.* (2021) introduced a MAUT-based method for selecting diplomatic representatives. The method was tested on a realistic dataset of 50 candidate diplomats. The algorithm's accuracy was 94%, and it was indicated that the method could serve as a reliable decision-support tool for strategic and bureaucratic appointments. To reduce uncertainties in choosing new branch locations for commercial businesses, Mesran et al. (2023) used an integrated ROC-MAUT method. The study identified the A6 alternative as the best location, proving that the ROC-MAUT combo is a flexible, practical, and effective tool for reducing error margins in business expansion decisions. For personnel performance measurement, Sulistiani et al. (2023) applied the PIPRECIA-S (Simplified Pivot Pairwise Relative Criteria Importance Assessment) method to weigh the criteria. These weights were combined with the MAUT approach to finalize the ranking of options. It was demonstrated that the combined PIPRECIA-S and MAUT method effectively handled different types of criteria in performance analysis and produced stable, comparable rankings. Addressing extreme decision environments, Zafar et al. (2026) focused on finding the best medical service robots (MSRs) for use during highly contagious epidemics like COVID-19. They used an entropy-based MAUT method based on spherical fuzzy set principles. Sensitivity and comparative tests showed that this method outperforms existing options in accuracy and efficiency. Overall, these applications indicate MAUT's adaptability across sensitive domains. They suggest that integrating MAUT with fuzzy logic and hybrid weighting tools may enhance decision consistency under conditions of uncertainty.

A review of the literature reveals that due to the broad and multidimensional nature of cybersecurity, the phenomenon has been studied extensively. However, it has been observed that the vast majority of existing studies focus on technical, legal, and theoretical frameworks, while research conducting performance evaluations specifically oriented around the GCI remains relatively limited. In index-based studies such as the NCSI and the GCI, member states of international platforms like the G7, G20, BRICS, and the European Union (EU) are generally examined; furthermore, MCDM analyses have remained comparatively scarce, with statistical approaches like hierarchical regression, correlation, and DEA being preferred instead. While these statistical approaches provide valuable descriptive statistics, they lack the capability to evaluate complex multi-attribute decision trade-offs inherent in cybersecurity frameworks. Traditional MCDM techniques often rely on subjective weighting (e.g., AHP), introducing expert bias, or struggle with non-linear utility trade-offs. Conversely, the MEREC method has found a wide spectrum of applications—including waste management, sensor selection, climate change impacts, and innovation performance analysis—due to its ability to derive objective weights through criterion removal effects without

expert intervention. Similarly, MAUT, as a contemporary method, has been utilized across diverse fields such as nuclear reactors, diplomatic representative selection, commercial branch site selection, and personnel performance measurement because of its strength in aggregating performance scores into a unified utility function. Accordingly, by attempting to address the methodological limitations of conventional approaches through the integration of MEREC and MAUT, this study seeks to contribute to the existing literature. It aims to offer a comprehensive cybersecurity evaluation within the scope of the SCO, which represents a prominent actor both regionally and globally.

3. Data and Methodology

This study aims to conduct an empirical evaluation of cybersecurity for a designated group of countries. Within this scope, GCI data, which serve as a highly comprehensive and significant metric in the field of cybersecurity, were analyzed using an integrated MCDM approach based on the MEREC MAUT integrated method. The sample of the study comprises the SCO member states and Türkiye.

The SCO constitutes a major block encompassing nations like China, India, and Russia, which together hold a significant share of the global population and economic output. It is an organization that accommodates critical countries with decisive roles in global cyber policies capable of shifting international balances. Originally established in 1996 under the name Shanghai Five by China, Russia, Kyrgyzstan, Tajikistan, and Kazakhstan, the organization aims to maintain regional peace, security, and stability, while jointly combating terrorism, extremism, separatism, organized crime, and illegal migration. Furthermore, it seeks to enhance cooperation in politics, economics, science and technology, culture and education, energy, and environmental issues (https://www.mfa.gov.tr/sanghay-isbirligi-orgutu.tr.mfa; Yılmaz; 2016).

Within the organization, which also includes observer and dialogue partner mechanisms, Türkiye was officially granted the status of a dialogue partner in 2012. Countries holding dialogue partner status have the opportunity to cooperate with the organization in specific domains. This status is positioned below observer status but above guest participant status. Crucially, this engagement operates at the intersection of economic integration and security-oriented functional cooperation. On the one hand, accelerating digital trade generate shared digital dependencies and mutual cyber vulnerabilities between Türkiye and SCO economies. On the other hand, cybersecurity constitutes a key domain within the SCO's institutional framework. Institutional channels, particularly those aligned with the Regional Anti-Terrorist Structure (RATS), facilitate strategic consultations on cyber-terrorism, critical information infrastructure protection, and regional data governance. The inclusion of Türkiye, which closely monitors the regional security vision of the organization and maintains its institutional and diplomatic interaction with the SCO, is of critical importance for evaluating the cybersecurity dynamics in the Eurasian geography through a holistic approach.

In addition, the fact that the member states are heterogeneous in terms of their cybersecurity infrastructures, legal regulations, and levels of technological development provides a methodologically suitable ground for MCDM methods to conduct performance comparisons. A review of the relevant literature reveals that the subject of cybersecurity has been extensively examined in the context of formations such as the EU, OECD, G7, and BRICS. Consequently, the present study is expected to bring a distinct perspective to the literature by identifying cybersecurity dynamics from a Eurasian viewpoint.

3.1. Data

The GCI was first introduced by the International Telecommunication Union (ITU) in 2015. This index aims to help countries identify their cybersecurity weaknesses and encourage them to strengthen their national capacity. It is a reliable reference that measures countries' global cybersecurity commitments and seeks to raise awareness of various aspects of the issue. Because cybersecurity is relevant across many industries, the development and participation levels of countries are analyzed using five core pillars. The overall index score for a country is calculated by combining scores in legal, technical, organizational, capacity, and cooperation categories. Since its inception in 2015 (v1), the GCI framework has evolved through updates in 2017 (v2), 2018 (v3), and 2020 (v4) to continuously refine its indicators and methodological accuracy

(Kravets, 2019). This study specifically leverages the GCI 2024 (v5) iteration to reflect current cybersecurity assessments.

GCI 2020 (v4) differs significantly from earlier versions as ICT became widespread alongside emerging areas like e-government, cloud computing, and the digital economy. In this version, legal measures were updated to align with national substantive law, and new indicators were added regarding privacy, harassment, and online safety. The technical measure CIRT (Computer Incident Response Team) was reorganized to better reflect operational principles. Crucially, moving from a declaration-based to an evidence-based approach, countries uploaded supporting documents and URLs to ensure accuracy. Additionally, the question set was expanded, and participating countries increased from 105 in 2015 to 150 in 2020 (ITU, 2020)**.**

The number of people with internet access has reached 5.4 billion by 2024. The integration of artificial intelligence, digitalization, and ongoing technological advances has also influenced GCI. A primary difference between the latest iteration, GCI 2024 (v5), and previous editions is the shift from the existing rank model to a tier-based GCI scoring system. In this model, a country's performance is categorized into five tiers, helping nations identify peer groups based on scores that they can use as benchmarks for development. Tier 1 (T1 - Role-models) includes countries with an overall score of at least 95 out of 100 and a strong government-led commitment across all five tiers or indicators. Tier 2 (T2 - Advancing) comprises countries scoring at least 85 points and having implemented cybersecurity measures, primarily through demonstrating strong determination, across up to four tiers. Countries in Tier 3 (T3 - Establishing) have scored at least 55 and show a basic level of government commitment to cybersecurity, with measures in only a limited number of areas or indicators. Tier 4 (T4 - Evolving) includes countries with at least 20 points, very weak infrastructure, and early steps taken in at least one area or a few sub-indicators. Tier 5 (T5 - Building) represents countries with fewer than 20 points, indicating the lowest cybersecurity level, with only basic improvements possible in at least one indicator or sub-indicator. To support these updates, the index questions have been clarified and improved, and the weighting recommendations have been revised according to the decisions of the GCI Expert Group. Additionally, data collection now involves a comprehensive, iterative review, analysis, revision, and approval process. Thanks to these quality checks, the reliability of the index data has been maximized (ITU, 2024).

The GCI consists of five pillars that are the fundamental building blocks of cybersecurity culture. These pillars include legal measures, technical measures, organizational measures, capacity development measures, and cooperation measures. The GCI questionnaire provides a score for the 20 indicators constructed through 83 questions. This fifth iteration of the GCI is scored on a scale of 0 to 100, with each pillar receiving 20 points. As a composite weighted index, each indicator, sub-indicator, and micro-indicator is assigned a weight based on its relative importance within the indicator group. Weighting recommendations are based on a budget allocation method. Experts were asked to contribute weighting recommendations for pillars in which they had expertise. The GCI was explicitly chosen as the primary benchmark over alternative international cybersecurity indices. This decision is grounded in its comprehensive five-pillar architecture and universal ITU standardization. Furthermore, its rigorous iterative data-verification mechanics align directly with our empirical decision-making framework. The empirical analysis relies on the GCI 2024 (v5) dataset, reflecting the latest officially available release at the time the study was conducted. The GCI 2024 (v5) pillars, indicators, associated codes, and optimization orientations (benefit/cost objectives) for 194 countries, along with their detailed explanations, are presented in Table 1 and the following paragraph (ITU, 2024).

Table 1. GCI 2024 Pillars and Indicators

Component		Code	Objective
Legal measures		**C1**	**max**
	Cybercrime laws Cybersecurity regulations		
Technical measures		**C2**	**max**
	National CERT/CIRT/CSIRT or SOC Sectoral CERT/CIRT/CSIRT or SOCs National framework for implementation of cybersecurity standards		
Organizational measures		**C3**	**max**
	National cybersecurity strategy Responsible agency Cybercrime metrics Child online protection strategies and initiatives		
Capacity development measures		**C4**	**max**
	Public cybersecurity awareness campaigns Training for cybersecurity professionals Cybersecurity educational programs in national academic curricula Cybersecurity research and development (R&D) programs National cybersecurity industry Government incentive mechanisms		
Cooperation measures		**C5**	**max**
	Bilateral cybersecurity agreements Multilateral cybersecurity agreements with other countries Mutual legal assistance treaties (MLATs) related to cybersecurity Public-private partnerships (PPPs) Inter-agency partnerships		

Source: ITU, 2024: 129-130.

Legal measures (C1): It assesses the presence of cybersecurity and cybercrime regulations in various countries. This pillar reflects the essential minimum of national cybersecurity capabilities, including the incorporation of cybercrime into criminal law, judicial procedures, enforcement of sanctions, and the development of institutional and legal frameworks for cyber governance.

Technical measures (C2) refer to institutional mechanisms that depend on technical institutions, standards, and frameworks for cybersecurity and combating cybercrime. Since the effective development and use of information and communication technologies (ICT) are only possible in a secure environment, these measures are based on a nation's technical capacity to identify cyber risks, prevent, respond to, and recover from cyber-attacks, as well as to improve overall national capabilities.

Organizational measures (C3): These involve national coordination organizations, policies, and strategic frameworks created to handle cybercrimes and boost cybersecurity. These include developing strategic plans and defining institutional roles and responsibilities, which are vital for establishing an effective national cybersecurity stance.

Capacity-development measures (C4): These refer to institutional mechanisms based on research and development (R&D) activities in cybersecurity, awareness-raising activities, training programs,

certification of experts, and public institutions that support cybersecurity development. These include social awareness campaigns, the certification and accreditation of experts, professional training, and the development of academic curricula. These capacity-building activities complement the success of the other three pillars of cybersecurity: legal, technical, and organizational measures.

Cooperation measures (C5): These refer to strategic structures based on partnerships, cooperation frameworks, and knowledge-sharing networks at the national, regional, and global levels. Due to digital interactions between states, cybersecurity has now become a shared global responsibility and an area of cross-border struggle. Stronger cooperation would enhance the cyber security capacities of countries and reduce cyber risks, while aiding in the identification, capture, and prosecution of malicious actors.

3.2. Methodology

In line with the objective of the research, the MEREC-based MAUT method is proposed for the analysis of the compiled GCI data. For determining the importance levels of the criteria, the MEREC method was preferred because it considers the removal effect of criteria and represents an objective approach, which is expected to produce more consistent weights. To integrate the criteria under a common utility function, MAUT—a contemporary MCDM method highly suitable for risk analysis—was selected. In this form, an objective, mathematically verifiable, and effective approach has been proposed. Detailed information regarding this proposed methodology is provided in the subsequent sections.

3.2.1. MEREC Method

An objective approach for criterion weighting, known as MEREC, was proposed by Keshavarz-Ghorabaee *et al.* (2021). The primary advantage of MEREC is its capacity to compute weights by individually omitting each criterion, which provides a transparent view of how such exclusions impact the remaining criteria. This distinctive feature sets MEREC apart from conventional objective weighting approaches, positioning it as a highly valuable tool for practical decision-making applications. MEREC offers significant advantages as an effective tool for handling complex decision-making problems by enhancing process transparency and flexibility. Unlike other complicate objective weighting methods, it provides a simpler, yet mathematically robust framework adaptable to diverse decision contexts (Ali, 2025). The implementation of MEREC for objective weight determination consists of the following steps (Ayçin, 2023; Ecer & Aycin, 2023; Keshavarz-Ghorabaee *et al.* 2021):

Step 1: A decision matrix is constructed as shown in Equation 1 (Eq(1)).

$$\left[X_{ij}\right]_{m\times n}=\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \; i=1,2,3,\dots m; \quad j=1,2,3,\dots n \tag{1}$$

Step 2: To standardize the decision matrix, its elements are subjected to a simple linear normalization in this step. The decision matrix is normalized using Eq. (2) for cost criteria and Eq. (3) for benefit criteria.

$$n_{ij}=\frac{x_{ij}}{\max\limits_{k} x_{kj}} \tag{2}$$

$$n_{ij}=\frac{\min\limits_{k} x_{kj}}{x_{ij}} \tag{3}$$

Step 3: In this step, a logarithmic measure with equal criteria weights is employed to determine the overall performances of the alternatives, and these calculations are carried out using Eq. (4).

$$S_i = \ln\left(1 + \left(\frac{1}{n}\sum_j \left|\ln\left(n_{ij}\right)\right|\right)\right) \tag{4}$$

Step 4: This step involves applying the logarithmic measure in a similar fashion to the preceding stage. The main difference from Step 3 lies in computing the alternatives' performances based on the separate removal of each criterion.

$$S'_{ij} = \ln\left(1 + \left(\frac{1}{n}\sum_{k,k\neq j} \left|\ln\left(n_{ij}\right)\right|\right)\right) \tag{5}$$

Step 5: The sum of the absolute deviations is calculated using Eq (6). Let E_j indicate the effect of eliminating the *j*th criterion.

$$E_j = \sum_i \left|S'_{ij} - S_i\right| \tag{6}$$

Step 6: The final weights of the criteria are determined via Eq (7). The weight of the *j*th criterion is represented by w_j.

$$w_j = \frac{E_j}{\sum_k E_k} \tag{7}$$

3.2.2. MAUT Method

The MAUT method was proposed by Keeney and Raiffa (1976). This approach is an extension of Multi-Attribute Value Theory (MAVT) and is grounded in the utility theory proposed by Von Neumann and Morgenstern (1947). In this methodology, a multi-attribute utility function is constructed to calculate the decision-maker's relative utility across the different alternatives. Owing to its robust foundation rooted in rational decision theory, MAUT captures the decision-maker's non-linear subjective preferences while directly integrating risk attitudes into the evaluation process under uncertainty. These attributes, encompassing theoretical soundness, flexibility, and risk modeling capabilities, constitute the prominent advantages of the method over other MCDM approaches. On account of these methodological advantages, MAUT was selected for integration with the MEREC method in a hybrid manner during the evaluation process. The implementation process of the proposed method consists of the following steps (Ecer, 2020; Løken & Botterud, 2005; Zhu et al., 2017):

Step 1: The decision-making matrix, which consists of m alternatives and n criteria, is structured as follows:

$$X = \left[X_{ij}\right]_{m\times n} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \; i = 1,2,3,\dots m; \;\; j = 1,2,3,\dots n \tag{8}$$

Step 2: The elements of the normalized decision matrix are derived using Eq. (9) and Eq. (10) for benefit and cost criteria, respectively

$$r_{ij} = \frac{x_{ij} - \min(x_{ij})}{\max(x_{ij}) - \min(x_{ij})} \quad (9)$$

$$r_{ij} = 1 + \frac{\min(x_{ij}) - x_{ij}}{\max(x_{ij}) - \min(x_{ij})} \quad (10)$$

Step 3: The marginal utility scores of the alternatives are computed using Eq. (11)

$$u_{ij} = \frac{e^{(r_{ij}^*)^2} - 1}{1{,}71} \quad (11)$$

where $e \cong 2{,}718$ represents the base of the natural logarithm.

Step 4: In this step, the ultimate utility score for each alternative is calculated via Eq. (12) by multiplying the marginal utility scores by their respective criterion weights.

$$U_i = \sum_{j=1}^{n} u_{ij} \bullet w_j \quad (12)$$

Step 5: Based on the ranking of the alternatives, the option yielding the highest score is selected as the best alternative.

Evaluating the cybersecurity performance of geopolitical entities like SCO member states involves multi-dimensional and non-linear risk factors under uncertainty. To address these complexities, the hybrid MEREC–MAUT framework was adopted based on specific methodological considerations.

As outlined in the preceding subsection regarding MEREC, its primary advantage lies in computing weights by individually omitting each criterion, showing how such exclusions might impact the remaining criteria. This feature sets MEREC apart from conventional objective approaches like Entropy, which purely measures statistical dispersion, or CRITIC, which relies on correlations. In cybersecurity, where omitting a single indicator (e.g., incident response) can impair the defense ecosystem, MEREC's removal-effect logic appears to reflect criterion criticality more effectively. Furthermore, unlike subjective tools like AHP that may introduce geopolitical bias, MEREC offers a simpler, mathematically robust framework adaptable to complex contexts (Ali, 2025), tending to provide a more neutral evaluation baseline.

Regarding MAUT, as noted above, it was selected for the alternative ranking stage due to its solid theoretical foundation in utility theory (Von Neumann & Morgenstern, 1947). MAUT constructs a multi-attribute utility function that captures non-linear preferences and risk attitudes under uncertainty. This offers potential advantages over distance-based methods like TOPSIS and VIKOR, which rely on linear Euclidean distances and can be sensitive to rank reversal or outlier distortions. MAUT is considered particularly suitable for modeling cybersecurity investments, where policy improvements often yield varying strategic utilities at different maturity stages.

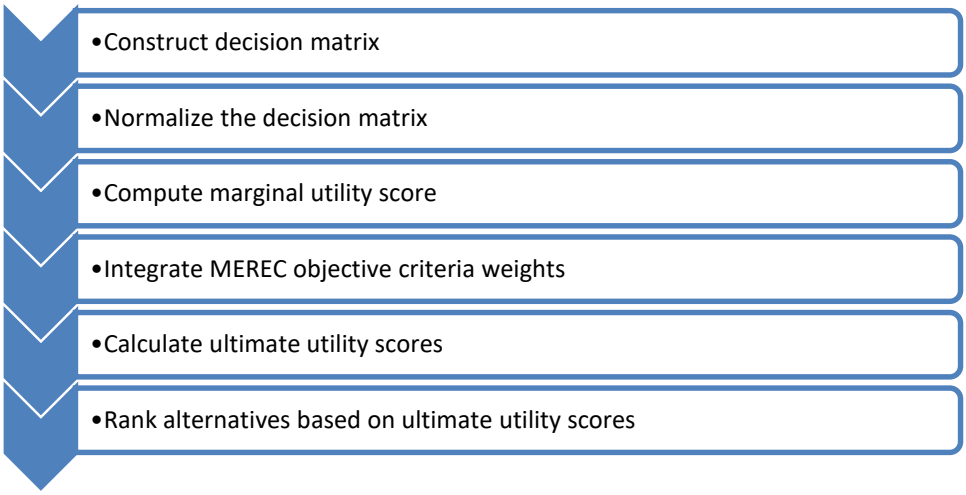
On account of these complementary attributes, MEREC helps minimize expert bias during weighting, while MAUT translates these weights into a risk-sensitive, utility-oriented evaluation suited for complex security environments.

4. Empirical Findings

The present study aimed to examine the cybersecurity performances of SCO countries and Türkiye. The GGI 2024 pillar data, calculated by ITU, were analyzed using the Hybrid MEREC-MAUT approach. Therefore, data from ten SCO member countries and Türkiye were used. The computational process and

decision-making framework applied in this evaluation are illustrated in Figure 1. The sequential implementation steps of the MEREC-weighted MAUT method are outlined below.

Figure 1. Flowchart of the MEREC-weighted MAUT Method



The GCI pillars included in the analysis were legal measures, technical measures, organizational measures, capacity-development measures, and cooperation measures, all of which served as decision criteria. The initial decision-making matrix is shown in Table 2.

Table 2. Initial Decision-Making Matrix

Country/Criteria	C1	C2	C3	C4	C5
China	20.00	17.14	18.34	18.46	17.70
Russian Fed.	20.00	16.59	20.00	18.77	16.77
Kyrgyzstan	16.71	14.87	16.32	5.53	12.16
Tajikistan	13.30	0.01	3.62	0.35	8.09
Kazakhstan	20.00	19.38	18.30	16.36	20.00
Uzbekistan	19.59	16.22	15.75	17.64	20.00
India	20.00	20.00	18.49	20.00	20.00
Pakistan	20.00	18.21	20.00	20.00	18.48
Iran	17.79	14.49	16.72	10.27	6.25
Belarus	18.61	2.78	11.97	8.98	19.11
Türkiye	20.00	20.00	20.00	20.00	20.00

Before presenting empirical findings and interpretations, the calculated descriptive statistics are included in Table 3.

Table 3. Descriptive Statistics for The Criterion Pillars

Criteria	Minimum	Maximum	Mean	Std. Deviation	Range
C1	13.30	20.00	18.73	2.12	6.70
C2	0.01	20.00	14.52	6.78	19.99
C3	3.62	20.00	16.32	4.84	16.38
C4	0.35	20.00	14.21	6.83	19.65
C5	6.25	20.00	16.23	5.05	13.75

The analysis of the data in Tables 2 and 3, based on the criteria and countries, showed that the mean and standard deviation of the legal measure (C1) figures did not vary significantly across countries. In contrast, technical measures (C2) displayed the opposite pattern. Tajikistan and Belarus scored well below

the mean, with high variation among countries. Regarding organizational measures (C3), the situation was similar, though the differences were not as pronounced as in technical measures (C2). Capacity-development measures (C4) in Tajikistan, Kyrgyzstan, Belarus, and Iran fell below the mean, while all other countries scored above average. Cooperation measures (C5) had generally high scores across most countries, although Tajikistan, Kyrgyzstan, and Iran had a mean score of six. Belarus scored high in this pillar. Since all pillar scores were high in Türkiye, the overall outlook appears quite positive. Overall, Türkiye, India, and Pakistan seem to be in good condition.

Before the MAUT analysis, the criterion weights were determined using MEREC, an objective weighting method. First, the initial decision-making matrix presented in Table 2 was normalized using Eq. (3), and both the normalized values and the Si values computed via Eq. (4) are provided in Table 4.

Table 4. Normalized Decision Matrix and Si Values

	Normalized decision matrix & Si					
Country/Criteria	**C 1**	**C 2**	**C 3**	**C4**	**C 5**	**Si**
China	0.665	0.001	0.197	0.019	0.353	1.360
Russian Fed	0.665	0.001	0.181	0.019	0.373	1.361
Kyrgyzstan	0.796	0.001	0.222	0.063	0.514	1.251
Tajikistan	1.000	1.000	1.000	1.000	0.773	0.050
Kazakhstan	0.665	0.001	0.198	0.021	0.313	1.366
Uzbekistan	0.679	0.001	0.230	0.020	0.313	1.352
India	0.665	0.001	0.196	0.018	0.313	1.379
Pakistan	0.665	0.001	0.181	0.018	0.338	1.374
Iran	0.748	0.001	0.217	0.034	1.000	1.252
Belarus	0.715	0.004	0.302	0.039	0.327	1.195
Türkiye	0.665	0.001	0.181	0.018	0.313	1.383

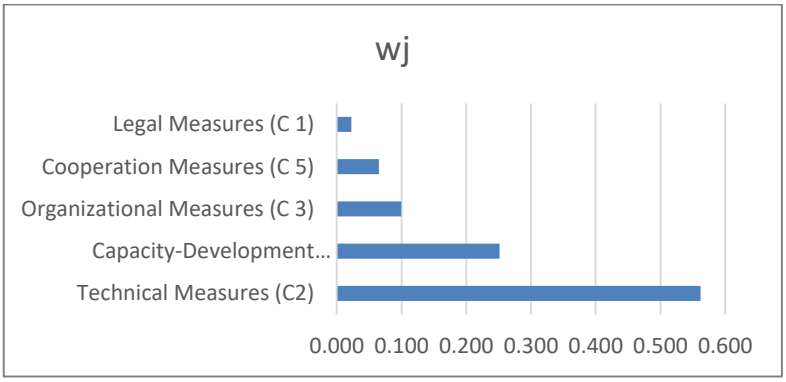
Following the calculation of Si values, the performance scores based on the individual removal of each criterion (S'ij) were computed using Eq. (5). Subsequently, the total absolute deviations (Ej) and the final objective criteria weights (wj) were calculated via Eq. (6) and Eq. (7), respectively. The resulting S'ij, Ej, and wj values are presented in Table 5.

Table 5. MEREC calculation Results (S'ij, Ej, and wj)

	Sij'	**Sij'**	**Sij'**	**Sij'**	**Sij'**
Country/Criteria	**C1**	**C2**	**C3**	**C4**	**C5**
China	1.339	0.879	1.273	1.133	1.305
Russian Fed	1.340	0.883	1.269	1.133	1.309
Kyrgyzstan	1.238	0.709	1.161	1.079	1.212
Tajikistan	0.050	0.050	0.050	0.050	0.000
Kazakhstan	1.345	0.878	1.280	1.148	1.305
Uzbekistan	1.332	0.871	1.273	1.126	1.290
India	1.358	0.896	1.293	1.151	1.318
Pakistan	1.353	0.896	1.283	1.145	1.317
Iran	1.235	0.713	1.160	1.037	1.252
Belarus	1.175	0.779	1.120	0.977	1.125
Türkiye	1.362	0.902	1.293	1.156	1.322
Ej	0.176	4.387	0.777	1.963	0.506
wj	**0.022**	**0.562**	**0.100**	**0.251**	**0.065**
Rank	5	1	3	2	4

As shown in Table 5, the highest weight is given to the technical measures (C2) pillar, followed by capacity development measures (C4) and organizational measures (C3). The lowest weight is assigned to the legal measures (C1) pillar. This criterion weighting and ranking pattern is visually illustrated in Figure 2.

Figure 2. Bar Chart



In the next step, all criteria were incorporated into the MAUT method as benefit-oriented for performance evaluation. Generally, as with all MCDM methods, the analysis begins by normalizing the decision-making matrix using Eq (9). The resulting decision-making matrix is shown in Table 6.

Table 6. Normalized Decision Matrix

	Normalized decision matrix				
Country/Criteria	**C1**	**C2**	**C3**	**C4**	**C5**
China	1.000	0.857	0.899	0.922	0.833
Russian Fed.	1.000	0.829	1.000	0.937	0.765
Kyrgyzstan	0.509	0.743	0.775	0.264	0.430
Tajikistan	0.000	0.000	0.000	0.000	0.134
Kazakhstan	1.000	0.969	0.896	0.815	1.000
Uzbekistan	0.939	0.811	0.741	0.880	1.000
India	1.000	1.000	0.908	1.000	1.000
Pakistan	1.000	0.910	1.000	1.000	0.889
Iran	0.670	0.724	0.800	0.505	0.000
Belarus	0.793	0.139	0.510	0.439	0.935
Türkiye	1.000	1.000	1.000	1.000	1.000

Using normalized decision matrix elements the marginal utility scores of the alternatives were calculated with Eq. (11) and are shown in Table 7.

Table 7. Marginal Utility Scores

	Marginal utility scores				
Country/Criteria	**C1**	**C2**	**C3**	**C4**	**C5**
China	1.005	0.634	0.727	0.783	0.585
Russian Fed.	1.005	0.579	1.005	0.823	0.465
Kyrgyzstan	0.173	0.431	0.482	0.042	0.119
Tajikistan	0.000	0.000	0.000	0.000	0.011
Kazakhstan	1.005	0.911	0.721	0.551	1.005
Uzbekistan	0.827	0.544	0.427	0.684	1.005
India	1.005	1.005	0.748	1.005	1.005
Pakistan	1.005	0.755	1.005	1.005	0.705
Iran	0.332	0.403	0.524	0.170	0.000
Belarus	0.511	0.011	0.174	0.124	0.818
Türkiye	1.005	1.005	1.005	1.005	1.005

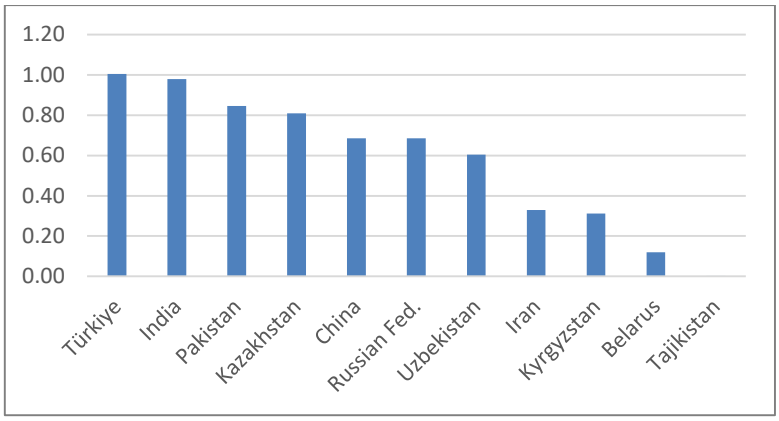
In the final MAUT step. marginal utility scores were multiplied by criterion weights. and the overall utility scores of the decision alternatives were calculated using Eq. (12) then presented in Table 8.

Table 8. Ultimate Utility Score

	Ultimate utility scores				
Country	**Ui**	**Rank**	**GCI tier**	**GCI tier score**	**GCI status**
China	0.69	5	T1	95-100	Role-modelling
Russian Fed.	0.68	6	T2	85-95	Advancing
Kyrgyzstan	0.31	9	T3	55-85	Establishing
Tajikistan	0.00	11	T4	20-55	Evolving
Kazakhstan	0.81	4	T2	85-95	Advancing
Uzbekistan	0.60	7	T2	85-95	Advancing
India	0.98	2	T1	95-100	Role-modelling
Pakistan	0.85	3	T1	95-100	Role-modelling
Iran	0.33	8	T3	55-85	Establishing
Belarus	0.12	10	T3	55-85	Establishing
Türkiye	1.00	1	T1	95-100	Role-modelling

As shown in Table 8 Türkiye which is not an SCO member ranked first in cybersecurity performance with the highest overall utility score, India, Pakistan, and Kazakhstan were the other top-performing countries. Conversely, Tajikistan performed significantly worse than the others in the group and ranked last. This is clearly evident in the column chart based on the ultimate utility score shown in Figure 3.

Figure 3. Column Chart



As shown in Fig. 3, China the Russian Federation and Uzbekistan are grouped in the moderate performance segment within the organization pillar, with similar overall utility scores. Similarly, Iran and Kyrgyzstan fall into the low-performance group, GCI tiers and scores presented in Table 6 support this conclusion.

Furthermore, sensitivity analyses were performed to measure the robustness of the applied MAUT method against changes in criteria weights. For this purpose, the weights obtained as a result of the MEREC, LOPCOW, Entropy, mean weight (MW), and standard deviation (SD) methods applied to the data as weighting techniques are presented in Table 9.

Table 9. Criterion Weights for Sensitivity Analysis

Weight/Criteria	C1	C2	C3	C4	C5
MEREC	0.022	0.562	0.100	0.251	0.065
LOPCOW	0.225	0.190	0.229	0.180	0.176
Entropy	0.015	0.381	0.128	0.352	0.125
MW	0.200	0.200	0.200	0.200	0.200
SD	0.068	0.280	0.178	0.288	0.187

Different MAUT rankings were generated using the weights obtained from the scenarios created through the five distinct objective weighting methods presented in Table 8. The resulting MAUT findings are provided in Table 10.

Table 10. Sensitivity Analysis Findings

	Ultimate utility scores									
	MEREC		LOPCOW		Entropy		MW		SD	
Country	Ui	Rank	Ui	Rank	Ui	Rank	Ui	Rank	Ui	Rank
China	0.69	5	0.76	6	0.70	6	0.75	6	0.71	6
Russian Fed.	0.68	6	0.80	5	0.71	5	0.78	5	0.73	5
Kyrgyzstan	0.31	9	0.26	10	0.26	9	0.25	10	0.25	10
Tajikistan	0.00	11	0.00	11	0.00	11	0.00	11	0.00	11
Kazakhstan	0.81	4	0.84	4	0.77	4	0.84	4	0.80	4
Uzbekistan	0.60	7	0.69	7	0.64	7	0.70	7	0.67	7
India	0.98	2	0.95	2	0.97	2	0.95	2	0.96	2
Pakistan	0.85	3	0.90	3	0.87	3	0.89	3	0.88	3
Iran	0.33	8	0.30	9	0.29	8	0.29	9	0.28	8
Belarus	0.12	10	0.32	8	0.18	10	0.33	8	0.26	9
Türkiye	1.00	1	1.00	1	1.00	1	1.00	1	1.00	1

When the MAUT rankings generated by the different criteria weights shared in Table 10 are examined, it is observed that Türkiye, India, Pakistan, Kazakhstan, Uzbekistan, and Tajikistan maintain the exact same rankings across all scenarios. On the other hand, the remaining countries exhibit minor shifts, moving either one step up or down within their respective categories. These results demonstrate that the methodology utilized in this research is robust against changes in criteria weights and generates highly consistent outcomes.

In the final stage of the study, alongside the MAUT method, the widely accepted TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) method was preferred. Distinct from the MAUT approach, TOPSIS performs its analysis based on a geometric distance logic that identifies the alternative closest to the positive ideal solution and farthest from the negative ideal solution. Therefore, assessing whether two methods with fundamentally different logics yield stable and consistent results is more appropriate for verifying the model's reliability. For this reason, the TOPSIS method, which is frequently employed in the literature for Cross-Method Validation, was selected. The empirical findings obtained at the end of this process are presented in Table 11.

Table 11. MAUT and TOPSIS Rankings

	MAUT					TOPSIS				
	MEREC	LOPCOW	Entropy	MW	SD	MEREC	LOPCOW	Entropy	MW	SD
Country	Rank	Rank	Rank	Rank	Rank	Rank	Rank	Rank	Rank	Rank
China	5	6	6	6	6	5	5	4	5	4
Russ. Fed.	6	5	5	5	5	6	6	6	6	6
Kyrgyzstan	9	10	9	10	10	9	9	9	9	9
Tajikistan	11	11	11	11	11	11	11	11	11	11
Kazakhstan	4	4	4	4	4	3	4	5	4	5
Uzbekistan	7	7	7	7	7	7	7	7	7	7
India	2	2	2	2	2	2	2	2	2	2
Pakistan	3	3	3	3	3	4	3	3	3	3
Iran	8	9	8	9	8	8	8	8	8	8
Belarus	10	8	10	8	9	10	10	10	10	10
Türkiye	1	1	1	1	1	1	1	1	1	1

The findings in Table 11 demonstrate that the methodology applied in this study yields outcomes consistent with the TOPSIS method, which is widely utilized in the literature. In all ten distinct rankings presented in Table 10, Türkiye, India, Uzbekistan, and Tajikistan occupy the exact same positions, while only minor method-based variations are observed in the rankings of the remaining countries. Furthermore, to evaluate the consistency of the generated rankings, the results of the Spearman rank correlation analysis are shared in Table 12.

Table 12. Spearman Correlation Analysis Findings for TOPSIS

		MAUT					TOPSIS				
		MEREC	LOPCOW	Entropy	MW	SD	MEREC	LOPCOW	Entropy	MW	SD
MAUT	MEREC	1.00									
	LOPCOW	0.96*	1.00								
	Entropy	0.99*	0.97*	1.00							
	MW	0.96*	1*	0.97*	1.00						
	SD	0.98*	0.99*	0.99*	0.99*	1.00					
TOPSIS	MEREC	0.99*	0.96*	0.98*	0.95*	0.97*	1.00				
	LOPCOW	1*	0.96*	0.99*	0.96*	0.98*	0.99*	1.00			
	Entropy	0.99*	0.95*	0.97*	0.95*	0.96*	0.97*	0.99*	1.00		
	MW	1*	0.96*	0.99*	0.96*	0.98*	0.99*	1*	0.99*	1.00	
	SD	0.99*	0.95*	0.97*	0.95*	0.96*	0.97*	0.99*	1*	0.99*	1.00

* $p<0.01$

The Spearman rank correlation coefficients provided in Table 12 are all statistically significant and positive. Consequently, these high correlation values indicate a remarkably strong relationship and consistency among the ten distinct rankings generated. This demonstrates the high internal stability, robustness, and intermethod agreement of the utilized MEREC-MAUT approach in conducting cybersecurity performance analyses.

To extend the cross-validation framework, the VIKOR method was additionally incorporated into the stability analysis. This addition aims to address the potential limitations of relying solely on distance-based techniques. VIKOR employs a linear normalization procedure, which differs from the vector normalization typically utilized in TOPSIS. Furthermore, it offers a compromise-ranking mechanism based on balancing maximum group utility and minimum individual regret. Evaluating alternatives through methods with distinct decision paradigms—namely utility-based MAUT, distance-based TOPSIS, and compromise-based VIKOR—is

considered beneficial. This multi-perspective approach tends to provide a more comprehensive assessment of model robustness. The empirical findings obtained from the VIKOR implementation across all weighting scenarios are presented in Table 13.

Table 13. MAUT and VIKOR Rankings

	MAUT					VIKOR				
	MEREC	**LOPCOW**	**Entropy**	**MW**	**SD**	**MEREC**	**LOPCOW**	**Entropy**	**MW**	**SD**
Country	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**	**Rank**
China	5	6	6	6	6	5	5	4	5	4
Russ. Fed.	6	5	5	5	5	6	6	6	6	6
Kyrgyzstan	9	10	9	10	10	9	8	9	8	9
Tajikistan	11	11	11	11	11	11	11	11	11	11
Kazakhstan	4	4	4	4	4	4	4	5	4	5
Uzbekistan	7	7	7	7	7	7	7	7	7	7
India	2	2	2	2	2	2	2	2	2	2
Pakistan	3	3	3	3	3	3	3	3	3	3
Iran	8	9	8	9	8	8	10	8	10	8
Belarus	10	8	10	8	9	10	9	10	9	10
Türkiye	1	1	1	1	1	1	1	1	1	1

Similarly, the findings in Table 13 indicate that the MAUT methodology yields results largely consistent with the VIKOR method. Across all ten distinct rankings presented in Table 13, Türkiye, Pakistan, India, Uzbekistan, and Tajikistan occupy the exact same positions. For the remaining countries, the observed differences appear to be limited to minor shifts of a single rank. These slight variations may be attributed to inherent methodological nuances between the two techniques. Additionally, to systematically evaluate the degree of consistency among the generated rankings, the results of the Spearman rank correlation analysis are presented in Table 14.

Table 14. Spearman Correlation Analysis Findings for VIKOR

	rho	**MAUT**					**VIKOR**				
		MEREC	**LOPCOW**	**Entropy**	**MW**	**SD**	**MEREC**	**LOPCOW**	**Entropy**	**MW**	**SD**
MAUT	**MEREC**	1.00									
	LOPCOW	0.96*	1.00								
	Entropy	0.99*	0.97*	1.00							
	MW	0.96*	1.00*	0.97*	1.00						
	SD	0.98*	0.99*	0.99*	0.99*	1.00					
VIKOR	**MEREC**	1.00*	0.96*	0.99*	0.96*	0.98*	1.00				
	LOPCOW	0.97*	0.96*	0.96*	0.96*	0.95*	0.97*	1.00			
	Entropy	0.99*	0.95*	0.97*	0.95*	0.96*	0.99*	0.96*	1.00		
	MW	0.97*	0.96*	0.96*	0.96*	0.96*	0.97*	1.00*	0.96*	1.00	
	SD	0.99*	0.95*	0.97*	0.95*	0.96*	0.99*	0.96*	1.00*	0.96*	1.00

* $p<0.01$

In line with the observations reported in Table 12, the Spearman rank correlation coefficients presented in Table 14 are all positive and statistically significant. These high correlation values suggest a strong degree of consistency among the ten generated rankings. Consequently, these empirical findings appear to support the internal stability, robustness, and intermethod agreement of the implemented MAUT-VIKOR framework in evaluating cybersecurity performance.

5. Conclusion

With the growth of digitalization, cyber threats have made it essential to increase awareness of cybersecurity and implement precautions. For countries, this is seen as a critical component of national security and economic stability. Assessing the cybersecurity performance of organizations with geopolitical and economic strategic importance, especially those like the SCO, is becoming increasingly important in the literature.

Accordingly, a quantitative evaluation approach based on the GCI dataset, a widely recognized international framework, was adopted in this study. The cybersecurity performance of SCO member countries and Türkiye was analyzed using the hybrid MEREC-MAUT approach. In the first stage, which included five criteria and 11 decision alternatives, the objective MEREC method was used to determine the criterion weights. According to the findings, technical measures (C2) ranked as the top criterion, followed by capacity-development measures (C4) and organizational measures (C3). These findings appear to align with Bruggemann et al. (2022) and Altıntaş (2022). In the second stage, performance rankings for the countries were determined using these weights and the MAUT method. Analysis results indicated that Türkiye, India, Pakistan, Kazakhstan, and China emerge as the top performers, whereas Tajikistan and Belarus exhibit lower performance levels by a noticeable margin. These observations are largely consistent with the report by Küpeli and Söğüt (2024). The analysis suggests that Türkiye displays a performance pattern comparable to top-tier nations. Its position among the leading group tends to highlight progress in legal and technical infrastructure, reflecting its developing capacity to respond to cyber incidents. Consequently, Türkiye may offer valuable insights as a potential reference model for other member states seeking to strengthen their institutional frameworks and craft strategic cyber defense policies.

Beyond presenting empirical rankings, the proposed framework seeks to offer notable methodological and practical insights. Methodologically, integrating objective MEREC weights with the flexible MAUT framework aims to mitigate subjective evaluation bias. By reducing reliance on expert-based scoring, this integration addresses a key analytical challenge under uncertainty. Furthermore, it accommodates potential non-linear utility functions and may serve as a adaptable template for complex security evaluations. The sensitivity analysis conducted in the final step of the study revealed no changes in the rankings of the countries with the highest and lowest performances, with variations observed only in a few countries shifting either one step up or down. This situation indicates that the approach applied in the research is robust against changes in criteria weights and generates stable results. Furthermore, a comparative analysis was performed via the TOPSIS and VIKOR methods, complemented by Spearman correlation analysis, to test the reliability of the proposed model's outputs. The findings demonstrate that the model yields dependable results and can be effectively utilized by researchers for cybersecurity performance evaluations. High-performing countries like Türkiye, India, and Pakistan serve as role models with high GCI scores, placing them in the T1 tier. This pattern also applies to other countries in the group. These nations excelled across GCI tier scores, technical measures (C2), capacity development measures (C4), and organizational measures (C3). As a result, MAUT findings, GCI scores, and tiering based on these scores all showed similar results. This consistency affirms the findings from the MAUT approach and highlights the value of coupling objective weights with utility-based ranking tools.

It would be advantageous for low-performing countries like Tajikistan and Belarus to quickly update their technical infrastructure and institutional mechanisms. They should focus on effective ICT utilization, boost R&D efforts, create policies in collaboration with relevant institutions to enhance cybersecurity, and support these initiatives with qualified training programs. Besides strategic investments, it would be wise to draw on the knowledge and experience of high-performing countries within the organization. To

operationalize these findings, policy makers in developing nations might consider leveraging the SCO Regional Anti-Terrorist Structure (RATS) framework. In this context, lagging nations could join a dedicated SCO Cyber Threat Intelligence Exchange Platform under RATS. Additionally, participating in a joint SCO Computer Emergency Response Team (SCO-CSIRT) network could enhance their technical response. This platform offers potential opportunities for joint threat intelligence and shared incident protocols. Additionally, establishing formal know-how transfer mechanisms with key regional actors, such as Türkiye, India, and Pakistan, could prove beneficial. To make this operational, nations could execute joint cyber defense drills with national Computer Emergency Response Teams (CERTs), such as Türkiye's National Cyber Incident Response Center (USOM). Furthermore, establishing a Critical Infrastructure Protection (CIP) Mentorship Program could help standardize technical resilience in key sectors.

The empirical scope of this study is primarily centered on 2024 data, focusing on SCO member states alongside Türkiye. Future research could extend the proposed framework to other international organizations, such as NATO, ASEAN, or the EU, in combination with alternative MCDM methodologies. Furthermore, incorporating dynamic multi-period datasets, additional indices like the NCSI, and relevant macro-variables could offer broader analytical insights and help enhance the generalizability of the findings.

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References

Ali, J. (2025). Copula extended power aggregation operators and MEREC method under linguistic q-rung orthopair fuzzy information: An application to plastic waste management solution selection. *Evolutionary Intelligence*, *18*(5), 96. https://doi.org/10.1007/s12065-025-01080-4

Alkan, M. (2012). *Siber güvenlik ve siber savaşlar*. TBMM İnternet Komisyonu. https://www.tbmm.gov.tr (Access Date: 02.06.2026).

Altıntaş, F. F. (2022). G7 Ülkelerinin siber güvenlik performanslarının analizi: Entropi tabanlı MABAC yöntemi ile bir uygulama. *Güvenlik Bilimleri Dergisi*, *11*(1), 263-286. https://doi.org/10.28956/gbd.1109776

Aslay, F. (2017). Siber saldırı yöntemleri ve Türkiye'nin siber güvenlik mevcut durum analizi. *International Journal of Multidisciplinary Studies and Innovative Technologies*, *1*(1), 24-28.

Ayçin, E. (2023). *Çok kriterli karar verme*. Nobel Yayınevi.

Bruggemann, R., Koppatz, P., Scholl, M., & Schuktomow, R. (2022). Global cybersecurity index (GCI) and the role of its 5 pillars. *Social Indicators Research*, *159*(1), 125-143. https://doi.org/10.1007/s11205-021-02739-y

Darıcılı, A. B. (2019). Türkiye'nin siber güvenlik politikalarının analizi: Türkiye'nin siber güvenlik modeli için öneriler. *TESAM Akademi Dergisi*, *6*(2), 11-33. https://doi.org/10.30626/tesamakademi.613517

Ecer, F. (2020). *Çok kriterli karar verme*. Seçkin Yayıncılık.

Ecer, F., & Aycin, E. (2023). Novel comprehensive MEREC weighting-based score aggregation model for measuring innovation performance: The case of G7 countries. *Informatica*, *34*(1), 53-83. https://doi.org/10.15388/22-INFOR494

Fagarazzi, A. (2024). Impact of digital development level on national cybersecurity index. *Business Excellence*, *18*(2), 37-62. https://doi.org/10.22598/pi-be/2024.18.2.xy

Goutam, R. K. (2015). Importance of cyber security. *International Journal of Computer Applications*, *111*(7), 14-17.

Hathaway, M. E. & Klimburg, A. (2012). Preliminary considerations: on national cyber security. A. Klimburg (Ed.), *National cyber security framework manual* (pp. 1-42). NATO CCD COE Publication.

International Telecommunication Union (ITU) (2026). Global Cybersecurity Index. https://www.itu.int/en/ITU-D/Cybersecurity/Pages/global-cybersecurity-index.aspx (Access Date: 19.05.2026).

ITU (2009). Serıes X: Data networks. open system communications and security telecommunication security: overview of cybersecurity. https://www.itu.int/rec/t-rec-x.1205-200804-i (Access Date: 02.06.2026).

ITU (2024). Global cyber security index. 5th edition. ITU Publication. https://www.itu.int/epublications/publication/global-cybersecurity-index-2024 (Access Date: 15.05.2026).

ITU (2020). Global cyber security index 2020. ITU Publication. https://www.itu.int/dms_pub/itu-d/opb/str/D-STR-GCI.01-2021-PDF-E.pdf (Access Date: 19.05.2026).

Keshavarz-Ghorabaee, M., Amiri, M., Zavadskas, E. K., Turskis, Z., & Antucheviciene, J. (2021). Determination of objective weights using a new method based on the removal effects of criteria (MEREC). *Symmetry*, *13*(4), 525. https://doi.org/10.3390/sym13040525

Kim, S. K., & Song, O. (2009). A MAUT approach for selecting a dismantling scenario for the thermal column in KRR-1. *Annals of Nuclear Energy*, *36*(2), 145-150. https://doi.org/10.1016/j.anucene.2008.11.034

Kravets, V. (2019). Comparative analysis of the cybersecurity indices and their applications. *Theoretical and Applied Cybersecurity*, *1*(1), 97-102. https://doi.org/10.20535/tacs.2664-29132019.1.169090

Küpeli, M., & Söğüt, E. (2024. October). Assessment of the cybersecurity efficiency of G20 countries using data envelopment analysis. In *2024 17th International Conference on Information Security and Cryptology (ISCTürkiye)* (pp. 1-4). IEEE. https://doi.org/10.1109/ISCTrkiye64784.2024.10779233

Løken, E., & Botterud, A. (2005). *Planning of mixed local energy distribution systems: A comparison of two multi-criteria decision methods*. 28th Annual IAEE International Conference, Taipei, Taiwan.

Maglaras, L. A., Kim, K. H., Janicke, H., Ferrag, M. A., Rallis, S., Fragkou, P., MAglaras, A., & Cruz, T. J. (2018). Cyber security of critical infrastructures. *ICT Express*, *4*(1), 42-45. https://doi.org/10.1016/j.icte.2018.02.001

Mesran, M., Kusuma, A. A., & Lubis, R. M. F. (2023). Decision support system for determining new branch location applying the MAUT method with ROC weighting. *Bulletin of Informatics and Data Science*, *2*(2), 67-76. http://dx.doi.org/10.61944/bids.v2i2.76

Nemure, T., Mupa, M. N., Agyei, K. G., Chisora, H. H., Mudzingwa, K., & Chiwanga, R. (2026). Cybersecurity of critical infrastructures: Challenges and future perspectives. *World Journal of Advanced Research and Reviews, 29*(03), 870-883. https://doi.org/10.30574/wjarr.2026.29.3.0621

Onumo, A., Cullen. A., & Ullah-Awan, I. (2017, August). An empirical study of cultural dimensions and cybersecurity development. In *2017 IEEE 5th International Conference on Future Internet of Things and Cloud (FiCloud)* (pp. 70-76). IEEE. https://doi.org/10.1109/FiCloud.2017.41

Öztaş, T. (2025). LOPCOW-MARCOS yaklaşımıyla BRICS ülkelerinde siber güvenlik değerlendirmesi. *Afyon Kocatepe Üniversitesi İktisadi ve İdari Bilimler Fakültesi Dergisi, 28*(1), 21-37. https://doi.org/10.33707/akuiibfd.1670493

Öztaş, G. Z. (2026). Avrupa Birliği'ne aday ülkelerin pandemi öncesi ve sonrası siber güvenlik performansı: Centroidous ve RAPS yöntemleri ile bir değerlendirme. *Black Sea Journal of Engineering and Science*, *9*(1), 415-432. https://doi.org/10.34248/bsengineering.1834925

Prasetya, A. W. Y., Suhardjo. B., & Munir. R. (2024). Regression analysis of the national cyber security index in the Southeast Asia Region. *Asian Journal of Social and Humanities*, *2*(5), 1206-1216. https://doi.org/10.59888/ajosh.v2i5.249

Ruhl, C., Hollis, D., Hoffman, W., & Maurer, T. (2022). *Cyberspace and geopolitics: Assessing global cybersecurity norm processes at a crossroads*. Carnegie Endowment for International Peace, Working paper.

Simic, V., Gokasar, I., Deveci. M., & Švadlenka, L. (2022). Mitigating climate change effects of urban transportation using a type-2 neutrosophic MEREC-MARCOS model. *IEEE Transactions on Engineering Management*, *71*, 3233-3249. https://doi.org/10.1109/TEM.2022.3207375

Sulistiani, H., Palupiningsih, P., Hamidy, F., Sari, P. L., & Khairunnisa, Y. (2023, November). Employee performance evaluation using multi-attribute utility theory (MAUT) with piprecia-s weighting: A case study in education institution. In *2023 International Conference on Informatics. Multimedia. Cyber and Informations System (ICIMCIS)* (pp. 369-373). IEEE. https://doi.org/10.1109/ICIMCIS60089.2023.10349017

Taufik, I., Alam, C. N., Mustofa, Z., Rusdiana, A., & Uriawan, W. (2021). Implementation of multi-attribute utility theory (MAUT) method for selecting diplomats. In *IOP Conference Series: Materials Science and Engineering* (Vol. 1098. No. 3. p. 032055). IOP Publishing. https://doi.org/10.1088/1757-899X/1098/3/032055

T.C. Dışişleri Bakanlığı (2026). Şangay İşbirliği Örgütü. https://www.mfa.gov.tr/sanghay-isbirligi-orgutu.tr.mfa (Access Date: 21.06.2026).

Wang, Q., Cheng, T., Lu, Y., Liu, H., Zhang, R., & Huang, J. (2024). Underground mine safety and health: A hybrid MEREC–CoCoSo system for the selection of best sensor. *Sensors*, *24*(4), 1285. https://doi.org/10.3390/s24041285

Yılmaz, Y. (2016). *Şanghay işbirliği örgütü adalet bakanları toplantısı ve Türkiye*. Uluslararası Hukuk ve Dış İlişkiler Genel Müdürlüğü, Uluslararası Hukuk Bülteni, 7, 5-6.

Zafar, F., Irfan, M., Rasheed, M. W., Fatima, S., Mahboob, A., & Sarwar, T. (2026). Optimizing medical service robot selection with spherical fuzzy rough MAUT approach. *Automatika*, *67*(1), 49-68. https://doi.org/10.1080/00051144.2025.2600153

Zhu, H., Ou, C. X., van den Heuvel, W. J. A., & Liu, H. (2017). Privacy calculus and its utility for personalization services in e-commerce: An analysis of consumer decision-making. *Information & Management*, *54*(4), 427-437. http://dx.doi.org/10.1016/j.im.2016.10.001